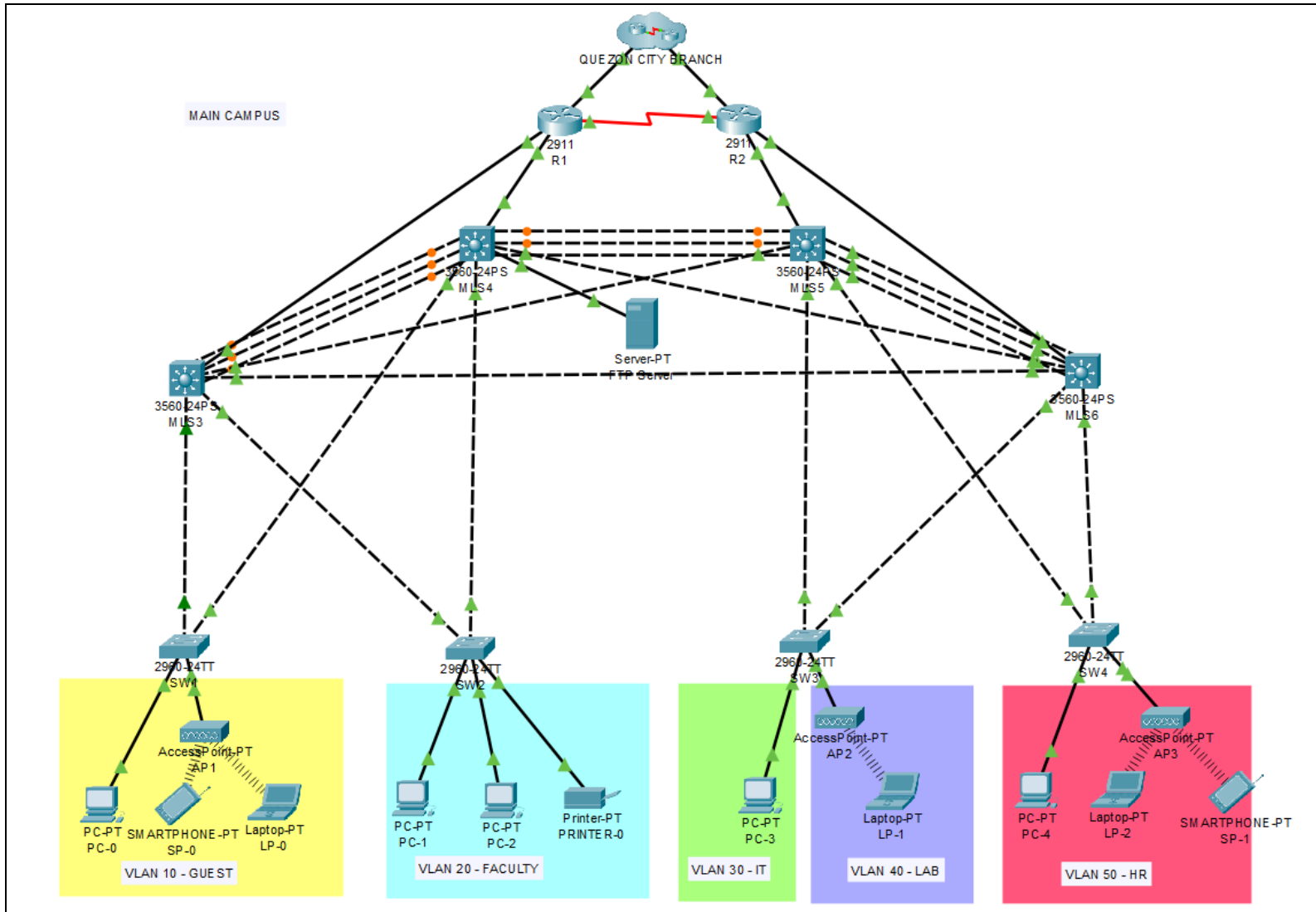


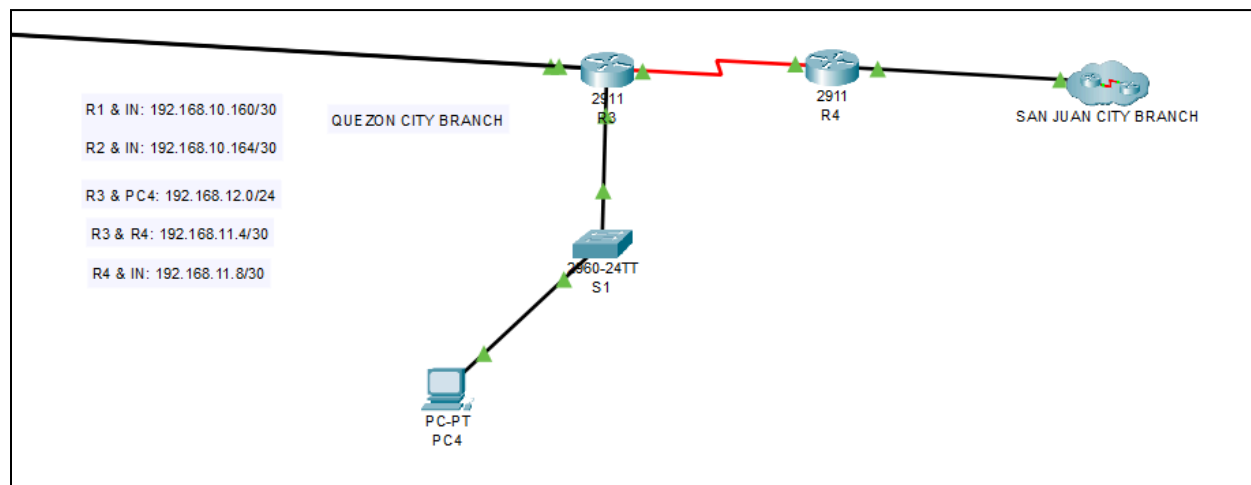
I. Network Diagram and Topology

WAN Topology:

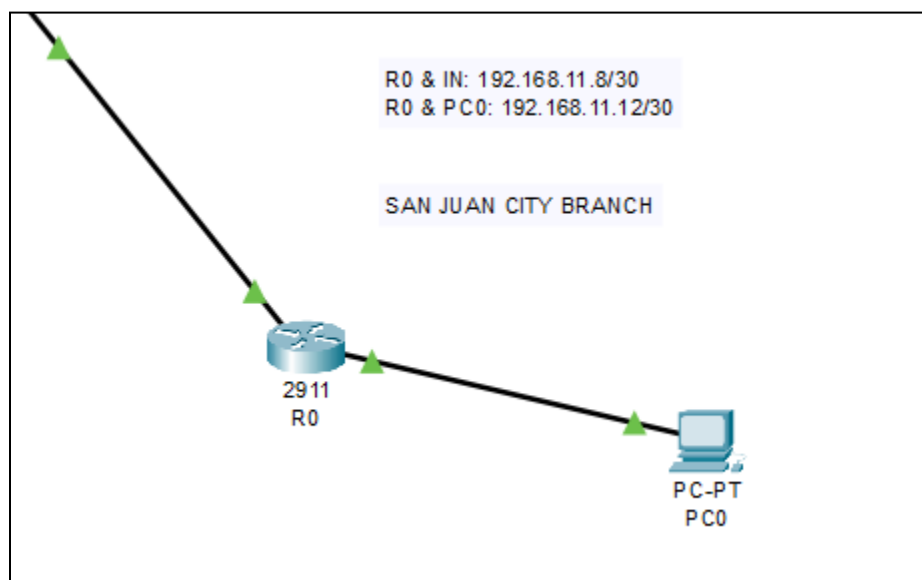
- PUP Main Campus:



- **PUP Quezon City Branch:**



II. PUP San Juan City Branch:



II. Introduction

The implementation of a multi-campus network infrastructure for the Polytechnic University of the Philippines (PUP) Sta. Mesa serves as the central hub connecting the university's branch campuses, including PUP Quezon City and PUP San Juan City, through a secure and scalable Wide Area Network (WAN) architecture. The network is designed to support 50 to 100 users per campus segment, enabling the deployment of essential network services such as file sharing via FTP, STP toolkits, inter-VLAN routing, dynamic IP addressing through

DHCP, and secure remote administration via SSH. Moreover, network security protocols such as port-security, DHCP Snooping, and extended ACLs for guests are also implemented to secure and protect the internal users and devices of the network. The main campus in Sta. Mesa houses the core infrastructure, consisting of dual core routers (R1 and R2), four multilayer switches (MLS3–MLS6) operating in a redundant distribution layer, and four access-layer switches (SW1–SW4) segmenting users into five VLANs: Guest (VLAN 10), Faculty (VLAN 20), IT Department (VLAN 30), Laboratories (VLAN 40), and Human Resources (VLAN 50). Branch campuses connect back to Sta. Mesa through dedicated routers and serial WAN links, ensuring consistent and reliable connectivity across all PUP campuses.

The goal of this network design is to provide fast, secure, and predictable connectivity while enforcing proper network segmentation, redundancy, and administrative control across all sites.

Addressing Table:

- **PUP Main Campus - Routers**

Device	Interface	IP Address	Subnet Mask	Network/Prefix
R1	G0/0	192.168.10.161	255.255.255.252	192.168.10.160/30
	G0/1	192.168.10.130	255.255.255.252	192.168.10.128/30
	G0/2	192.168.10.138	255.255.255.252	192.168.10.136/30
	Se0/3/1	192.168.10.145	255.255.255.252	192.168.10.144/30
R2	G0/0	192.168.10.165	255.255.255.252	192.168.10.164/30
	G0/1	192.168.10.150	255.255.255.252	192.168.10.148/30
	G0/2	192.168.10.158	255.255.255.252	192.168.10.156/30
	Se0/3/1	192.168.10.146	255.255.255.252	192.168.10.144/30

- **PUP Main Campus - Multilayer Switches (Routed Ports & Port-Channels)**

Device	Interface	IP Address	Subnet Mask	Network/Prefix
MLS3	Po1	192.168.10.113	255.255.255.252	192.168.10.112/30
	Fa0/1	192.168.10.129	255.255.255.252	192.168.10.128/30
	Fa0/17	192.168.10.125	255.255.255.252	192.168.10.124/30
	Fa0/20	192.168.10.141	255.255.255.252	192.168.10.140/30
	Fa0/2	192.168.10.133	255.255.255.252	192.168.10.132/30
	VLAN 10 (SVI)	192.168.10.2	255.255.255.224	192.168.10.0/27
	VLAN 20 (SVI)	192.168.10.34	255.255.255.224	192.168.10.32/27
	Loopback1	1.1.1.1	255.255.255.255	—
	Po1	192.168.10.114	255.255.255.252	192.168.10.112/30
	Po3	192.168.10.121	255.255.255.252	192.168.10.120/30

Device	Interface	IP Address	Subnet Mask	Network/Prefix
MLS4	Fa0/1	192.168.10.153	255.255.255.252	192.168.10.152/30
	Fa0/2	192.168.10.137	255.255.255.252	192.168.10.136/30
	Fa0/18	192.168.10.133	255.255.255.252	192.168.10.132/30
	VLAN 10 (SVI)	192.168.10.3	255.255.255.224	192.168.10.0/27
	VLAN 20 (SVI)	192.168.10.35	255.255.255.224	192.168.10.32/27
	Loopback2	2.2.2.2	255.255.255.255	—
MLS5	Po2	192.168.10.117	255.255.255.252	192.168.10.116/30
	Po3	192.168.10.122	255.255.255.252	192.168.10.120/30
	Fa0/4	192.168.10.149	255.255.255.252	192.168.10.148/30
	Fa0/17	192.168.10.126	255.255.255.252	192.168.10.124/30
	VLAN 30 (SVI)	192.168.10.66	255.255.255.240	192.168.10.64/28
	VLAN 40 (SVI)	192.168.10.82	255.255.255.240	192.168.10.80/28
	VLAN 50 (SVI)	192.168.10.98	255.255.255.240	192.168.10.96/28
	VLAN 100 (SVI)	192.168.10.52	255.255.255.252	—
	Loopback3	3.3.3.3	255.255.255.255	—
MLS6	Po2	192.168.10.118	255.255.255.252	192.168.10.116/30
	Fa0/3	192.168.10.157	255.255.255.252	192.168.10.156/30
	Fa0/18	192.168.10.134	255.255.255.252	192.168.10.132/30
	Fa0/20	192.168.10.142	255.255.255.252	192.168.10.140/30
	VLAN 30 (SVI)	192.168.10.67	255.255.255.240	192.168.10.64/28
	VLAN 40 (SVI)	192.168.10.83	255.255.255.240	192.168.10.80/28
	VLAN 50 (SVI)	192.168.10.99	255.255.255.240	192.168.10.96/28
	VLAN 100 (SVI)	192.168.10.53	255.255.255.252	—
	Loopback4	4.4.4.4	255.255.255.255	—

- PUP Main Campus - FTP Server & End Devices

Device	VLAN	IP Address	Subnet Mask	Network/Prefix
FTP Server	VLAN 100	192.168.10.151	255.255.255.252	—
PC-0	VLAN 10	DHCP	255.255.255.224	192.168.10.2 / .3
SP-0	VLAN 10	DHCP	255.255.255.224	192.168.10.2 / .3
LP-0	VLAN 10	DHCP	255.255.255.224	192.168.10.2 / .3
PC-1	VLAN 20	DHCP	255.255.255.224	192.168.10.34 / .35
PC-2	VLAN 20	DHCP	255.255.255.224	192.168.10.34 / .35
Printer-0	VLAN 20	DHCP	255.255.255.224	192.168.10.34 / .35
PC-3	VLAN 30	DHCP	255.255.255.240	192.168.10.66 / .67
LP-1	VLAN 40	DHCP	255.255.255.240	192.168.10.82 / .83
PC-4	VLAN 50	DHCP	255.255.255.240	192.168.10.98 / .99
LP-2	VLAN 50	DHCP	255.255.255.240	192.168.10.98 / .99
SP-1	VLAN 50	DHCP	255.255.255.240	192.168.10.98 / .99

- PUP Quezon City Branch - Routers

Device	Interface	IP Address	Subnet Mask	Network/Prefix
R3	G0/0	192.168.10.162	255.255.255.252	192.168.10.160/30
	G0/1	192.168.10.166	255.255.255.252	192.168.10.164/30
	G0/2	192.168.12.1	255.255.255.0	192.168.12.0/24
	Se0/3/0	192.168.11.5	255.255.255.252	192.168.11.4/30
	Loopback1	1.1.1.1	255.255.255.255	—
	Loopback9	9.9.9.9	255.255.255.255	—

Device	Interface	IP Address	Subnet Mask	Network/Prefix
R4	G0/2	192.168.11.10	255.255.255.252	192.168.11.8/30
	Se0/3/0	192.168.11.6	255.255.255.252	192.168.11.4/30
	Loopback1	1.1.1.1	255.255.255.255	—
	Loopback9	9.9.9.9	255.255.255.255	—
PC4 (QC)	—	DHCP	255.255.255.0	192.168.12.1

- **PUP San Juan City - Routers**

Device	Interface	IP Address	Subnet Mask	Network/Prefix
R0	G0/0	192.168.11.9	255.255.255.252	192.168.11.8/30
	G0/2	192.168.11.13	255.255.255.252	192.168.11.12/30
PC0 (Pasig)	—	DHCP	255.255.255.252	192.168.11.13

III. Design Considerations

Network Fundamentals

- Hierarchical three-layer network design (core/distribution/access) to simplify deployment, troubleshooting, and management across all PUP campuses.

IP Addressing and VLAN Design

- Subnetting from the 192.168.10.0/24 network to allocate appropriately sized subnets per VLAN and per point-to-point link.
- Private IP addressing used throughout all campuses with /27, /28, /30 prefix lengths depending on host requirements.
- Dynamic IP address allocation via DHCP for end devices, with static assignments reserved for servers, SVIs, and routed ports.
- Inter-VLAN routing handled at the distribution layer by MLS3–MLS6, eliminating the need for router-on-a-stick, improving network performance.
- Separation of management traffic using dedicated VLAN 100 (native/management VLAN) on distribution switches.

Switching Technologies (LAN Layer 2 Design)

- EtherChannel (Port-Channel) links between distribution layer switches for increased bandwidth and redundancy.
- Rapid recovery in case of component failure through redundant distribution-layer paths between MLS3–MLS6.
- VLAN segmentation enforcing logical separation between Guest, Faculty, IT, Laboratories, and HR users.
- Disabling of unused ports and interfaces on all switches and routers to prevent unauthorized access.

LAN Services

- The access layer provides Fast Ethernet connectivity to end devices through SW1–SW4 and S1.
- Spanning Tree Protocol (STP) enhancements, including PortFast, were configured on access ports to optimize convergence and improve network efficiency for end-user connections.
- Wireless access provided through Access Points (AP1, AP2, AP3) at SW1, SW3, and SW4 supporting mobile end devices.
- Centralized DHCP services are provided by the multilayer switches (MLS), enabling automatic IP address assignment for all VLANs to simplify network administration and improve scalability.
- Internal users are granted full access to network services and resources, while guest users are restricted to limited connectivity, specifically access to the FTP server only, to ensure network security and segmentation.
- Provisions for accommodating future VLAN expansion beyond the current five VLANs (Guest, Faculty, IT, Laboratories, HR).

WAN Services

- Point-to-point WAN links connecting PUP Sta. Mesa (main campus) to PUP Quezon City and PUP Pasig branches, providing dedicated bandwidth for data and administrative traffic.
- Redundant WAN connections between R1 and R2 (Se0/3/1) and R3 and R4 were designed to provide failover capability and improve network reliability. The topology was adapted based on available router interfaces in the simulation environment.
- Point-to-point /30 subnets across all WAN links to conserve IP address space.
- Automatic switchover to backup WAN link via the redundant serial connection between main campus and QC branch.

IP Routing and Layer 3 Design

- OSPFv2 is implemented in a single-area design to provide dynamic routing and ensure efficient exchange of routing information across all network segments.

- Complementary static routing is configured for selected routes requiring fixed path control and administrative preference.
- Loopback interfaces configured on all multilayer switches and branch routers for stable router ID assignment and management reachability.
- Distribution layer implemented using four multilayer switches (MLS3–MLS6) operating at Layer 3 with inter-VLAN routing capabilities.

High Availability and Redundancy

- High availability through redundant routers (R1 and R2) at the main campus core, both connected to branch routers R3 and R4.
- HSRP (Hot Standby Router Protocol) is deployed to provide gateway redundancy by enabling automatic failover of the default gateway, ensuring uninterrupted connectivity for end devices during router downtime or failure scenarios.
- Near-wire-speed performance between all devices through Layer 3 switching at the distribution layer.

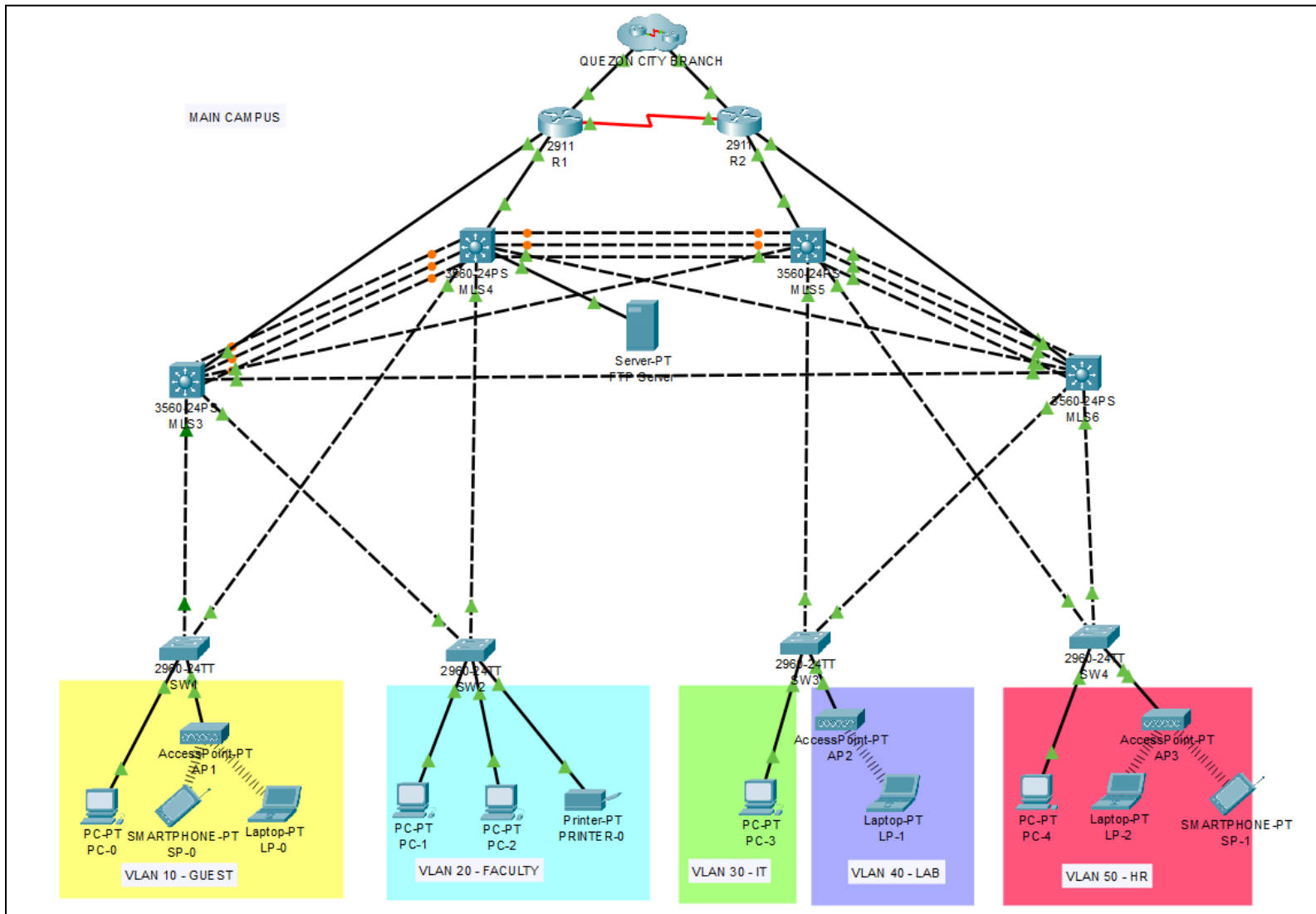
Security Services

- SSH enabled on all network devices for secure remote administration, replacing insecure Telnet access.
- Authentication required for all remote management sessions via SSH with configured credentials.
- Access control is enforced at the distribution layer to restrict inter-VLAN communication where necessary.
- Logging and auditing of network device access for accountability and troubleshooting across all three campuses.
- Physical security considerations for networking equipment housed at PUP Sta. Mesa main campus server/network room.

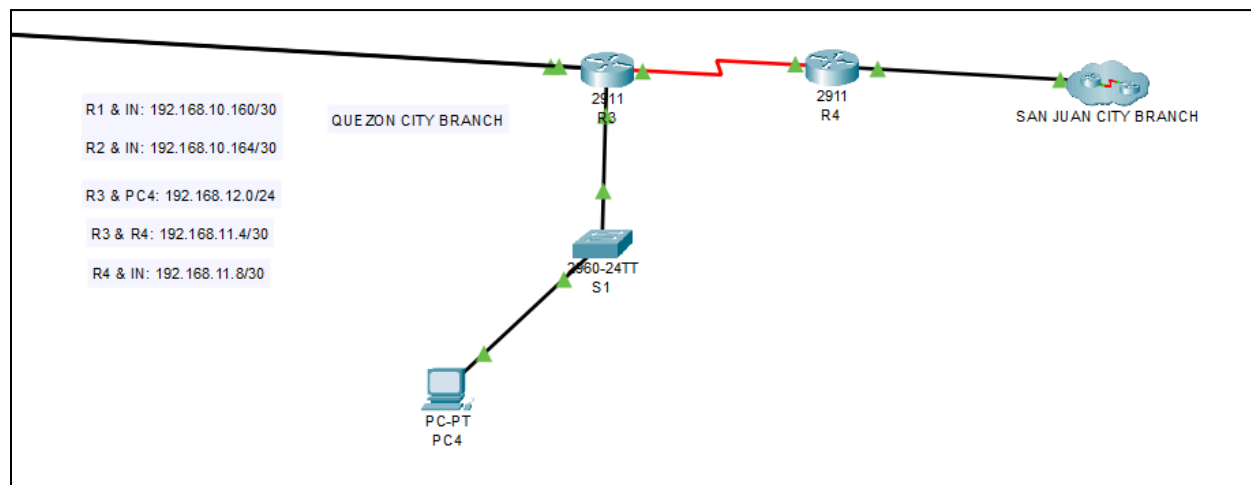
IV. Network Diagrams and Topologies

WAN Topology:

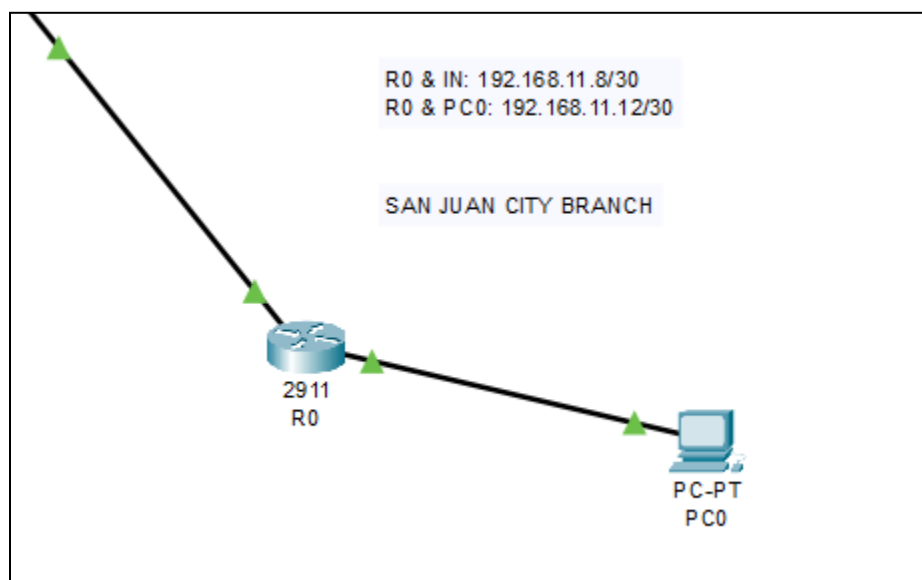
- PUP Main Campus:



- **PUP Quezon City Branch:**



- **PUP San Juan City Branch:**



V. System Components

Device	Platform	Modules
R1	Cisco 2911 ISR Router	Serial WAN module (Se0/3/1), 3x GigabitEthernet (G0/0–G0/2)
R2	Cisco 2911 ISR Router	Serial WAN module (Se0/3/1), 3x GigabitEthernet (G0/0–G0/2)
R3	Cisco 2911 ISR Router	Serial WAN module (Se0/3/0), 3x GigabitEthernet (G0/0–G0/2)
R4	Cisco 2911 ISR Router	Serial WAN module (Se0/3/0), 3x GigabitEthernet (G0/0–G0/2)
R0	Cisco 2911 ISR Router	2x GigabitEthernet (G0/0, G0/2)
MLS3	Cisco 3560-24PS Multilayer Switch	24x FastEthernet, 2x GigabitEthernet, PoE, Layer 3 capable
MLS4	Cisco 3560-24PS Multilayer Switch	24x FastEthernet, 2x GigabitEthernet, PoE, Layer 3 capable
MLS5	Cisco 3560-24PS Multilayer Switch	24x FastEthernet, 2x GigabitEthernet, PoE, Layer 3 capable
MLS6	Cisco 3560-24PS Multilayer Switch	24x FastEthernet, 2x GigabitEthernet, PoE, Layer 3 capable
SW1	Cisco 2960-24TT Access Switch	24x FastEthernet, 2x GigabitEthernet uplink
SW2	Cisco 2960-24TT Access Switch	24x FastEthernet, 2x GigabitEthernet uplink

Device	Platform	Modules
SW3	Cisco 2960-24TT Access Switch	24x FastEthernet, 2x GigabitEthernet uplink
SW4	Cisco 2960-24TT Access Switch	24x FastEthernet, 2x GigabitEthernet uplink
S1	Cisco 2960-24TT Access Switch	24x FastEthernet, 2x GigabitEthernet uplink
AP1	Cisco AccessPoint-PT	Wireless access for VLAN 10 (Guest) at SW1
AP2	Cisco AccessPoint-PT	Wireless access for VLAN 30 (IT) at SW3
AP3	Cisco AccessPoint-PT	Wireless access for VLAN 50 (HR) at SW4
FTP Server	Server-PT	FTP service, static IP, connected to MLS4/MLS5 via VLAN 100

VI. Configurations

SW1
<pre>! version 15.0 no service timestamps log datetime msec no service timestamps debug datetime msec service password-encryption ! hostname SW1 ! enable secret 5 \$1\$mERr\$AFX/pZT1Lh7NP3Dp3P/qq/ ! ! ip dhcp excluded-address 192.168.10.3 ! ip ssh version 2 ip ssh time-out 60 ip domain-name school.edu ! username Admin secret 5 \$1\$mERr\$AFX/pZT1Lh7NP3Dp3P/qq/ ! ! ip dhcp snooping vlan 10 no ip dhcp snooping information option ip dhcp snooping ! spanning-tree mode pvst spanning-tree portfast default spanning-tree extend system-id ! interface FastEthernet0/1 switchport access vlan 999 switchport mode access shutdown ! interface FastEthernet0/2 switchport access vlan 999 switchport mode access shutdown ! interface FastEthernet0/3 switchport access vlan 999</pre>

SW1

```
switchport mode access
shutdown
!
interface FastEthernet0/4
switchport access vlan 999
switchport mode access
shutdown
!
interface FastEthernet0/5
switchport access vlan 999
switchport mode access
shutdown
!
interface FastEthernet0/6
switchport access vlan 999
switchport mode access
shutdown
!
interface FastEthernet0/7
switchport access vlan 999
switchport mode access
shutdown
!
interface FastEthernet0/8
switchport access vlan 999
switchport mode access
shutdown
!
interface FastEthernet0/9
switchport access vlan 999
switchport mode access
shutdown
!
interface FastEthernet0/10
switchport access vlan 999
switchport mode access
shutdown
!
interface FastEthernet0/11
switchport access vlan 999
switchport mode access
shutdown
!
```


SW1

```
interface FastEthernet0/12
switchport access vlan 999
switchport mode access
shutdown
!
interface FastEthernet0/13
switchport access vlan 999
switchport mode access
shutdown
!
interface FastEthernet0/14
switchport access vlan 999
switchport mode access
shutdown
!
interface FastEthernet0/15
switchport access vlan 999
switchport mode access
shutdown
!
interface FastEthernet0/16
switchport access vlan 999
switchport mode access
shutdown
!
interface FastEthernet0/17
switchport access vlan 999
switchport mode access
shutdown
!
interface FastEthernet0/18
switchport access vlan 999
switchport mode access
shutdown
!
interface FastEthernet0/19
switchport access vlan 999
switchport mode access
shutdown
!
interface FastEthernet0/20
switchport access vlan 999
switchport mode access
```

SW1

```
shutdown
!
interface FastEthernet0/21
switchport access vlan 999
switchport mode access
shutdown
!
interface FastEthernet0/22
switchport access vlan 999
switchport mode access
shutdown
!
interface FastEthernet0/23
description PC (DEV) 1st Floor
switchport access vlan 10
ip dhcp snooping limit rate 15
switchport mode access
switchport port-security
switchport port-security mac-address sticky
switchport port-security violation restrict
switchport port-security mac-address sticky 0001.6357.299C
!
interface FastEthernet0/24
description AP (WLAN) from 1st Floor
switchport access vlan 10
ip dhcp snooping limit rate 15
switchport mode access
switchport port-security
switchport port-security maximum 2
switchport port-security mac-address sticky
switchport port-security violation restrict
switchport port-security mac-address sticky 0001.632A.92EA
switchport port-security mac-address sticky 0002.4AB2.5DDB
!
interface GigabitEthernet0/1
switchport trunk allowed vlan 10
ip dhcp snooping trust
switchport mode trunk
!
interface GigabitEthernet0/2
switchport trunk allowed vlan 10
ip dhcp snooping trust
switchport mode trunk
```

SW1

```

!
interface Vlan1
no ip address
shutdown
!
banner motd ^C
*****
WARNING: Authorized Access Only
This device is property of the School Network.
Unauthorized access is strictly prohibited.
*****
^C
!
!
!
line con 0
password 7 082048430017544541
logging synchronous
login local
!
line vty 0 4
login local
transport input ssh
line vty 5 15
login local
transport input ssh
!
!
!
!
end

```

SW2

```

!
version 15.0
no service timestamps log datetime msec
no service timestamps debug datetime msec
service password-encryption
!

```

SW2

```
hostname SW2
!
enable secret 5 $1$mERr$AFX/pZT1Lh7NP3Dp3P/qq/
!
!
!
ip ssh version 2
ip ssh time-out 60
ip domain-name school.edu
!
username Admin secret 5 $1$mERr$AFX/pZT1Lh7NP3Dp3P/qq/
!
!
ip dhcp snooping vlan 20,100
no ip dhcp snooping information option
ip dhcp snooping
!
spanning-tree mode pvst
spanning-tree portfast default
spanning-tree extend system-id
!
interface FastEthernet0/1
switchport access vlan 999
switchport mode access
shutdown
!
interface FastEthernet0/2
switchport access vlan 999
switchport mode access
shutdown
!
interface FastEthernet0/3
switchport access vlan 999
switchport mode access
shutdown
SW2#sh run
Building configuration...
Current configuration : 4168 bytes
!
version 15.0
no service timestamps log datetime msec
no service timestamps debug datetime msec
service password-encryption
```

SW2

```
!  
hostname SW2  
!  
enable secret 5 $1$mERr$AFX/pZT1Lh7NP3Dp3P/qq/  
!  
!  
!  
ip ssh version 2  
ip ssh time-out 60  
ip domain-name school.edu  
!  
username Admin secret 5 $1$mERr$AFX/pZT1Lh7NP3Dp3P/qq/  
!  
!  
ip dhcp snooping vlan 20,100  
no ip dhcp snooping information option  
ip dhcp snooping  
!  
spanning-tree mode pvst  
spanning-tree portfast default  
spanning-tree extend system-id  
!  
interface FastEthernet0/1  
switchport access vlan 999  
switchport mode access  
shutdown  
!  
interface FastEthernet0/2  
switchport access vlan 999  
switchport mode access  
shutdown  
!  
interface FastEthernet0/3  
switchport access vlan 999  
switchport mode access  
shutdown  
!  
interface FastEthernet0/4  
switchport access vlan 999  
switchport mode access  
shutdown  
!  
interface FastEthernet0/5
```

SW2

```
switchport access vlan 999
switchport mode access
shutdown
!
interface FastEthernet0/6
switchport access vlan 999
switchport mode access
shutdown
!
interface FastEthernet0/7
switchport access vlan 999
switchport mode access
shutdown
!
interface FastEthernet0/8
switchport access vlan 999
switchport mode access
shutdown
!
interface FastEthernet0/9
switchport access vlan 999
switchport mode access
shutdown
!
interface FastEthernet0/10
switchport access vlan 999
switchport mode access
shutdown
!
interface FastEthernet0/11
switchport access vlan 999
switchport mode access
shutdown
!
interface FastEthernet0/12
switchport access vlan 999
switchport mode access
shutdown
!
interface FastEthernet0/13
switchport access vlan 999
switchport mode access
shutdown
```

SW2

```
!  
interface FastEthernet0/14  
switchport access vlan 999  
switchport mode access  
shutdown  
!  
interface FastEthernet0/15  
switchport access vlan 999  
switchport mode access  
shutdown  
!  
interface FastEthernet0/16  
switchport access vlan 999  
switchport mode access  
shutdown  
!  
interface FastEthernet0/17  
switchport access vlan 999  
switchport mode access  
shutdown  
!  
interface FastEthernet0/18  
switchport access vlan 999  
switchport mode access  
shutdown  
!  
interface FastEthernet0/19  
switchport access vlan 999  
switchport mode access  
shutdown  
!  
interface FastEthernet0/20  
switchport access vlan 999  
switchport mode access  
shutdown  
!  
interface FastEthernet0/21  
switchport access vlan 999  
switchport mode access  
shutdown  
!  
interface FastEthernet0/22  
switchport access vlan 20
```

SW2

```

ip dhcp snooping limit rate 15
switchport mode access
switchport port-security
switchport port-security mac-address sticky
switchport port-security violation restrict
switchport port-security mac-address sticky 0000.0CD5.BA15
!
interface FastEthernet0/23
description VoIP and PC (MARKETING) from 2nd Floor
switchport access vlan 20
ip dhcp snooping limit rate 15
switchport mode access
switchport voice vlan 100
switchport port-security
switchport port-security maximum 2
switchport port-security mac-address sticky
switchport port-security violation restrict
switchport port-security mac-address sticky 0001.433E.11E7
mls qos trust dscp
!
interface FastEthernet0/24
description Printer (MARKETING) from 2nd Floor
switchport access vlan 20
ip dhcp snooping limit rate 15
switchport mode access
switchport port-security
switchport port-security mac-address sticky
switchport port-security violation restrict
switchport port-security mac-address sticky 00E0.8FEE.4299
!
interface GigabitEthernet0/1
switchport trunk allowed vlan 20,100
ip dhcp snooping trust
switchport mode trunk
!
interface GigabitEthernet0/2
switchport trunk allowed vlan 20,100
ip dhcp snooping trust
switchport mode trunk
!
interface Vlan1
no ip address
shutdown

```


SW2

```

!
banner motd ^C
*****
WARNING: Authorized Access Only
This device is property of the School Network.
Unauthorized access is strictly prohibited.
*****
^C
!
!
!
line con 0
password 7 082048430017544541
logging synchronous
login local
!
line vty 0 4
login local
transport input ssh
line vty 5 15
login local
transport input ssh
!
!
!
!
end

```

SW3

```

!
version 15.0
no service timestamps log datetime msec
no service timestamps debug datetime msec
service password-encryption
!
hostname SW3
!
enable secret 5 $1$mERr$AFX/pZT1Lh7NP3Dp3P/qq/
!
!

```

SW3

```
!  
ip ssh version 2  
ip ssh time-out 60  
ip domain-name school.edu  
!  
username Admin secret 5 $1$mERr$AFX/pZT1Lh7NP3Dp3P/qq/  
!  
!  
ip dhcp snooping vlan 30,40  
no ip dhcp snooping information option  
ip dhcp snooping  
!  
spanning-tree mode pvst  
spanning-tree portfast default  
spanning-tree extend system-id  
!  
interface FastEthernet0/1  
switchport access vlan 999  
switchport mode access  
shutdown  
!  
interface FastEthernet0/2  
switchport access vlan 999  
switchport mode access  
shutdown  
!  
interface FastEthernet0/3  
switchport access vlan 999  
switchport mode access  
shutdown  
!  
interface FastEthernet0/4  
switchport access vlan 999  
switchport mode access  
shutdown  
!  
interface FastEthernet0/5  
switchport access vlan 999  
switchport mode access  
shutdown  
!  
interface FastEthernet0/6  
switchport access vlan 999
```

SW3

```
switchport mode access
shutdown
!
interface FastEthernet0/7
switchport access vlan 999
switchport mode access
shutdown
!
interface FastEthernet0/8
switchport access vlan 999
switchport mode access
shutdown
!
interface FastEthernet0/9
switchport access vlan 999
switchport mode access
shutdown
!
interface FastEthernet0/10
switchport access vlan 999
switchport mode access
shutdown
!
interface FastEthernet0/11
switchport access vlan 999
switchport mode access
shutdown
!
interface FastEthernet0/12
switchport access vlan 999
switchport mode access
shutdown
!
interface FastEthernet0/13
switchport access vlan 999
switchport mode access
shutdown
!
interface FastEthernet0/14
switchport access vlan 999
switchport mode access
shutdown
!
```

SW3

```
interface FastEthernet0/15
switchport access vlan 999
switchport mode access
shutdown
!
interface FastEthernet0/16
switchport access vlan 999
switchport mode access
shutdown
!
interface FastEthernet0/17
switchport access vlan 999
switchport mode access
shutdown
!
interface FastEthernet0/18
switchport access vlan 999
switchport mode access
shutdown
!
interface FastEthernet0/19
switchport access vlan 999
switchport mode access
shutdown
!
interface FastEthernet0/20
switchport access vlan 999
switchport mode access
shutdown
!
interface FastEthernet0/21
switchport access vlan 999
switchport mode access
shutdown
!
interface FastEthernet0/22
switchport access vlan 999
switchport mode access
shutdown
!
interface FastEthernet0/23
switchport access vlan 40
ip dhcp snooping limit rate 15
```

SW3

```

switchport mode access
switchport port-security
switchport port-security mac-address sticky
switchport port-security violation restrict
switchport port-security mac-address sticky 0030.F21D.C7E3
!
interface FastEthernet0/24
switchport access vlan 30
ip dhcp snooping limit rate 15
switchport mode access
switchport port-security
switchport port-security mac-address sticky
switchport port-security violation restrict
switchport port-security mac-address sticky 0090.0C19.A7AD
!
interface GigabitEthernet0/1
switchport trunk allowed vlan 30,40
ip dhcp snooping trust
switchport mode trunk
!
interface GigabitEthernet0/2
switchport trunk allowed vlan 30,40
ip dhcp snooping trust
switchport mode trunk
!
interface Vlan1
no ip address
shutdown
!
interface Vlan90
ip address 192.168.10.5 255.255.255.240
standby 90 ip 192.168.10.1
standby 90 priority 110
standby 90 preempt
!
banner motd ^C
*****
WARNING: Authorized Access Only
This device is property of the School Network.
Unauthorized access is strictly prohibited.
*****
^C
!
```

SW3

```
!  
!  
line con 0  
password 7 082048430017544541  
logging synchronous  
login local  
!  
line vty 0 4  
login local  
transport input ssh  
line vty 5 15  
login local  
transport input ssh  
!  
!  
!  
!  
end
```

SW4

```
Current configuration : 3965 bytes  
!  
version 15.0  
no service timestamps log datetime msec  
no service timestamps debug datetime msec  
service password-encryption  
!  
hostname SW4  
!  
enable secret 5 $1$mERr$AFX/pZT1Lh7NP3Dp3P/qq/  
!  
!  
!  
ip ssh version 2  
ip ssh time-out 60  
ip domain-name school.edu  
!  
username Admin secret 5 $1$mERr$AFX/pZT1Lh7NP3Dp3P/qq/
```

SW4

```
!  
!  
ip dhcp snooping vlan 50,100  
no ip dhcp snooping information option  
ip dhcp snooping  
!  
spanning-tree mode pvst  
spanning-tree portfast default  
spanning-tree extend system-id  
!  
interface FastEthernet0/1  
switchport access vlan 999  
switchport mode access  
shutdown  
!  
interface FastEthernet0/2  
switchport access vlan 999  
switchport mode access  
shutdown  
!  
interface FastEthernet0/3  
switchport access vlan 999  
switchport mode access  
shutdown  
!  
interface FastEthernet0/4  
switchport access vlan 999  
switchport mode access  
shutdown  
!  
interface FastEthernet0/5  
switchport access vlan 999  
switchport mode access  
shutdown  
!  
interface FastEthernet0/6  
switchport access vlan 999  
switchport mode access  
shutdown  
!  
interface FastEthernet0/7  
switchport access vlan 999  
switchport mode access
```

SW4

```
shutdown
!  
interface FastEthernet0/8  
switchport access vlan 999  
switchport mode access  
shutdown  
!  
interface FastEthernet0/9  
switchport access vlan 999  
switchport mode access  
shutdown  
!  
interface FastEthernet0/10  
switchport access vlan 999  
switchport mode access  
shutdown  
!  
interface FastEthernet0/11  
switchport access vlan 999  
switchport mode access  
shutdown  
!  
interface FastEthernet0/12  
switchport access vlan 999  
switchport mode access  
shutdown  
!  
interface FastEthernet0/13  
switchport access vlan 999  
switchport mode access  
shutdown  
!  
interface FastEthernet0/14  
switchport access vlan 999  
switchport mode access  
shutdown  
!  
interface FastEthernet0/15  
switchport access vlan 999  
switchport mode access  
shutdown  
!  
interface FastEthernet0/16
```


SW4

```
switchport access vlan 999
switchport mode access
shutdown
!
interface FastEthernet0/17
switchport access vlan 999
switchport mode access
shutdown
!
interface FastEthernet0/18
switchport access vlan 999
switchport mode access
shutdown
!
interface FastEthernet0/19
switchport access vlan 999
switchport mode access
shutdown
!
interface FastEthernet0/20
switchport access vlan 999
switchport mode access
shutdown
!
interface FastEthernet0/21
switchport access vlan 999
switchport mode access
shutdown
!
interface FastEthernet0/22
switchport access vlan 999
switchport mode access
shutdown
!
interface FastEthernet0/23
switchport access vlan 50
ip dhcp snooping limit rate 15
switchport mode access
switchport port-security
switchport port-security maximum 2
switchport port-security mac-address sticky
switchport port-security violation restrict
switchport port-security mac-address sticky 0060.3E99.0E76
```

SW4

```

switchport port-security mac-address sticky 0060.4746.6E5E
!
interface FastEthernet0/24
switchport access vlan 50
ip dhcp snooping limit rate 15
switchport mode access
switchport voice vlan 100
switchport port-security
switchport port-security maximum 2
switchport port-security mac-address sticky
switchport port-security violation restrict
switchport port-security mac-address sticky 0001.4391.59C2
mls qos trust dscp
!
interface GigabitEthernet0/1
switchport trunk allowed vlan 50,100
ip dhcp snooping trust
switchport mode trunk
!
interface GigabitEthernet0/2
switchport trunk allowed vlan 50,100
ip dhcp snooping trust
switchport mode trunk
!
interface Vlan1
no ip address
shutdown
!
banner motd ^C
*****
WARNING: Authorized Access Only
This device is property of the School Network.
Unauthorized access is strictly prohibited.
*****
^C
!
!
!
line con 0
password 7 082048430017544541
logging synchronous
login local
!

```

SW4

```
line vty 0 4
login local
transport input ssh
line vty 5 15
login local
transport input ssh
!
!
!
!
end
```

R1

```
!
version 15.1
no service timestamps log datetime msec
no service timestamps debug datetime msec
service password-encryption
!

hostname R1
!
enable secret 5 $1$mERr$AFX/pZT1Lh7NP3Dp3P/qq/
!
no ip cef
no ipv6 cef
!
username Admin secret 5 $1$mERr$AFX/pZT1Lh7NP3Dp3P/qq/
!
!
license udi pid CISCO2911/K9 sn FTX1524Y637-
!
ip ssh version 2
ip ssh time-out 60
ip domain-name school.edu
!
!
spanning-tree mode pvst
!
```

R1

```
interface GigabitEthernet0/0
ip address 192.168.10.161 255.255.255.252
duplex auto
speed auto
!
interface GigabitEthernet0/1
ip address 192.168.10.130 255.255.255.252
duplex auto
speed auto
!
interface GigabitEthernet0/2
ip address 192.168.10.138 255.255.255.252
duplex auto
speed auto
Current configuration : 1742 bytes
!
version 15.1
no service timestamps log datetime msec
no service timestamps debug datetime msec
service password-encryption

!
interface GigabitEthernet0/1/0
no ip address
shutdown
!
interface Serial0/3/0
no ip address
clock rate 2000000
shutdown
!
interface Serial0/3/1
ip address 192.168.10.145 255.255.255.252
clock rate 2000000
!
interface Vlan1
no ip address
shutdown
!
router ospf 1
log-adjacency-changes
network 192.168.10.128 0.0.0.3 area 0
network 192.168.10.136 0.0.0.3 area 0
```

R1

```

network 192.168.10.144 0.0.0.3 area 0
default-information originate
!
ip classless
ip route 0.0.0.0 0.0.0.0 192.168.10.162
!
ip flow-export version 9
!
banner motd ^C
*****
WARNING: Authorized Access Only
This device is property of the School Network.
Unauthorized access is strictly prohibited.
*****
^C
!
line con 0
password 7 082048430017544541
logging synchronous
login local
!
line aux 0
!
line vty 0 4
login local
transport input ssh
line vty 5 15
login local
transport input ssh
!
end

```

R2

```

Current configuration : 1742 bytes
!
version 15.1
no service timestamps log datetime msec
no service timestamps debug datetime msec
service password-encryption

```

R2

```
!  
hostname R2  
!  
enable secret 5 $1$mERr$AFX/pZT1Lh7NP3Dp3P/qq/  
!  
no ip cef  
no ipv6 cef  
!  
username Admin secret 5 $1$mERr$AFX/pZT1Lh7NP3Dp3P/qq/  
!  
!  
license udi pid CISCO2911/K9 sn FTX1524PS30-  
!  
ip ssh version 2  
ip ssh time-out 60  
ip domain-name school.edu  
!  
!  
spanning-tree mode pvst  
!  
interface GigabitEthernet0/0  
ip address 192.168.10.165 255.255.255.252  
duplex auto  
speed auto  
!  
interface GigabitEthernet0/1  
ip address 192.168.10.150 255.255.255.252  
duplex auto  
speed auto  
!  
interface GigabitEthernet0/2  
ip address 192.168.10.158 255.255.255.252  
duplex auto  
speed auto  
!  
interface Serial0/3/0  
no ip address  
clock rate 2000000  
shutdown  
!  
interface Serial0/3/1  
ip address 192.168.10.146 255.255.255.252  
!
```

R2

```

interface Vlan1
no ip address
shutdown
!
router ospf 1
log-adjacency-changes
network 192.168.10.144 0.0.0.3 area 0
network 192.168.10.148 0.0.0.3 area 0
network 192.168.10.156 0.0.0.3 area 0
default-information originate
!
ip classless
ip route 0.0.0.0 0.0.0.0 192.168.10.166
!
ip flow-export version 9
!
banner motd ^C
*****
WARNING: Authorized Access Only
This device is property of the School Network.
Unauthorized access is strictly prohibited.
*****
^C
!
line con 0
password 7 082048430017544541
logging synchronous
login local
!
line aux 0
!
line vty 0 4
login local
transport input ssh
line vty 5 15
login local
transport input ssh
!
end

```

R3

Current configuration : 1742 bytes

```
!  
version 15.1  
no service timestamps log datetime msec  
no service timestamps debug datetime msec  
service password-encryption  
  
!  
hostname R3  
!  
no ip cef  
no ipv6 cef  
!  
!  
!  
!  
license udi pid CISCO2911/K9 sn FTX152404TI-  
!  
spanning-tree mode pvst  
!  
interface Loopback1  
ip address 1.1.1.1 255.255.255.255  
!  
interface Loopback9  
ip address 9.9.9.9 255.255.255.255  
!  
interface GigabitEthernet0/0  
ip address 192.168.10.162 255.255.255.252  
duplex auto  
speed auto  
!  
interface GigabitEthernet0/1  
ip address 192.168.10.166 255.255.255.252  
duplex auto  
speed auto  
!  
interface GigabitEthernet0/2  
ip address 192.168.12.1 255.255.255.0  
duplex auto  
speed auto  
!  
interface Serial0/3/0  
ip address 192.168.11.5 255.255.255.252
```


R3

```
clock rate 2000000
!  
interface Serial0/3/1  
no ip address  
clock rate 2000000  
shutdown  
!  
interface Vlan1  
no ip address  
shutdown  
!  
router ospf 1  
log-adjacency-changes  
passive-interface GigabitEthernet0/2  
network 192.168.10.160 0.0.0.3 area 0  
network 192.168.10.164 0.0.0.3 area 0  
!  
ip classless  
ip route 0.0.0.0 0.0.0.0 192.168.10.161  
ip route 0.0.0.0 0.0.0.0 192.168.10.165  
ip route 192.168.11.0 255.255.255.252 192.168.11.6  
ip route 192.168.11.8 255.255.255.252 192.168.11.6  
ip route 192.168.11.12 255.255.255.252 192.168.11.6  
!  
ip flow-export version 9  
!  
line con 0  
!  
line aux 0  
!  
line vty 0 4  
login  
!  
end
```

R4

```
Current configuration : 1742 bytes  
!  
version 15.1
```

R4

```
no service timestamps log datetime msec
no service timestamps debug datetime msec
service password-encryption

!
hostname R4
!
no ip cef
no ipv6 cef
!
!
!
!
license udi pid CISCO2911/K9 sn FTX1524RG9U-
!
spanning-tree mode pvst
!
interface Loopback1
ip address 1.1.1.1 255.255.255.255
!
interface Loopback9
ip address 9.9.9.9 255.255.255.255
!
interface GigabitEthernet0/0
ip address 192.168.10.162 255.255.255.252
duplex auto
speed auto
!
interface GigabitEthernet0/1
no ip address
duplex auto
speed auto
!
interface GigabitEthernet0/2
ip address 192.168.11.10 255.255.255.252
duplex auto
speed auto
!
interface Serial0/3/0
ip address 192.168.11.6 255.255.255.252
!
interface Serial0/3/1
no ip address
```

R4

```
clock rate 2000000
shutdown
!
interface Vlan1
no ip address
shutdown
!
router ospf 1
log-adjacency-changes
passive-interface GigabitEthernet0/2
network 192.168.10.160 0.0.0.3 area 0
network 192.168.10.164 0.0.0.3 area 0
!
ip classless
ip route 0.0.0.0 0.0.0.0 192.168.10.161
ip route 0.0.0.0 0.0.0.0 192.168.10.165
ip route 0.0.0.0 0.0.0.0 192.168.11.5
ip route 192.168.11.12 255.255.255.252 192.168.11.9
ip route 192.168.12.0 255.255.255.0 192.168.11.5
!
ip flow-export version 9
!

line con 0
!
line aux 0
!
line vty 0 4
login
!
end
```

R0

```
Current configuration : 1742 bytes
!
version 15.1
no service timestamps log datetime msec
no service timestamps debug datetime msec
service password-encryption
```

R0

```
!  
hostname R0  
!  
ip cef  
no ipv6 cef  
!  
license udi pid CISCO2911/K9 sn FTX1524M6NP-  
!  
spanning-tree mode pvst  
!  
interface GigabitEthernet0/0  
ip address 192.168.11.9 255.255.255.252  
duplex auto  
speed auto  
!  
interface GigabitEthernet0/1  
no ip address  
duplex auto  
speed auto  
shutdown  
!  
interface GigabitEthernet0/2  
ip address 192.168.11.13 255.255.255.252  
duplex auto  
speed auto  
!  
interface Vlan1  
no ip address  
shutdown  
!  
ip classless  
ip route 192.168.11.4 255.255.255.252 192.168.11.10  
ip route 0.0.0.0 0.0.0.0 192.168.11.10  
ip route 192.168.12.0 255.255.255.0 192.168.11.5  
ip route 192.168.12.0 255.255.255.0 192.168.11.10  
!  
ip flow-export version 9  
!  
line con 0  
!  
line aux 0  
!  
line vty 0 4
```

R0
<pre>login ! end</pre>

MLS3
<pre>Current configuration : 5300 bytes ! version 12.2(37)SE1 no service timestamps log datetime msec no service timestamps debug datetime msec service password-encryption ! hostname MLS3 ! ! enable secret 5 \$1\$mERr\$AFX/pZT1Lh7NP3Dp3P/qq/ ! ! ip dhcp excluded-address 192.168.10.1 192.168.10.2 ip dhcp excluded-address 192.168.10.33 192.168.10.34 ip dhcp excluded-address 192.168.10.2 ip dhcp excluded-address 192.168.10.34 ip dhcp excluded-address 192.168.10.3 ! ip dhcp pool VLAN10 network 192.168.10.0 255.255.255.224 default-router 192.168.10.1 dns-server 8.8.8.8 ip dhcp pool VLAN20 network 192.168.10.32 255.255.255.224 default-router 192.168.10.33 dns-server 8.8.8.8 ! ! ip routing ! !</pre>

MLS3

```
!  
!  
username Admin secret 5 $1$mERr$AFX/pZT1Lh7NP3Dp3P/qq/  
!  
!  
!  
!  
!  
!  
!  
!  
!  
!  
!  
ip ssh version 2  
ip ssh time-out 60  
ip domain-name school.edu  
!  
!  
spanning-tree mode pvst  
!  
!  
!  
!  
!  
!  
interface Loopback1  
ip address 1.1.1.1 255.255.255.255  
!  
interface Port-channel1  
no switchport  
ip address 192.168.10.113 255.255.255.252  
!  
interface FastEthernet0/1  
no switchport  
ip address 192.168.10.129 255.255.255.252  
duplex auto  
speed auto  
!  
interface FastEthernet0/2  
no switchport  
ip address 192.168.10.133 255.255.255.252  
duplex auto  
speed auto
```

MLS3

```
!  
interface FastEthernet0/3  
switchport access vlan 999  
switchport mode access  
shutdown  
!  
interface FastEthernet0/4  
switchport access vlan 999  
switchport mode access  
shutdown  
!  
interface FastEthernet0/5  
switchport access vlan 999  
switchport mode access  
shutdown  
!  
interface FastEthernet0/6  
switchport access vlan 999  
switchport mode access  
shutdown  
!  
interface FastEthernet0/7  
switchport access vlan 999  
switchport mode access  
shutdown  
!  
interface FastEthernet0/8  
switchport access vlan 999  
switchport mode access  
shutdown  
!  
interface FastEthernet0/9  
switchport access vlan 999  
switchport mode access  
shutdown  
!  
interface FastEthernet0/10  
switchport access vlan 999  
switchport mode access  
shutdown  
!  
interface FastEthernet0/11  
switchport access vlan 999
```

MLS3

```
switchport mode access
shutdown
!
interface FastEthernet0/12
switchport access vlan 999
switchport mode access
shutdown
!
interface FastEthernet0/13
switchport access vlan 999
switchport mode access
shutdown
!
interface FastEthernet0/14
switchport access vlan 999
switchport mode access
shutdown
!
interface FastEthernet0/15
switchport access vlan 999
switchport mode access
shutdown
!
interface FastEthernet0/16
switchport access vlan 999
switchport mode access
shutdown
!
interface FastEthernet0/17
no switchport
ip address 192.168.10.125 255.255.255.252
duplex auto
speed auto
!
interface FastEthernet0/18
switchport access vlan 999
switchport mode access
shutdown
!
interface FastEthernet0/19
switchport access vlan 999
switchport mode access
shutdown
```


MLS3

```
!  
interface FastEthernet0/20  
no switchport  
ip address 192.168.10.141 255.255.255.252  
duplex auto  
speed auto  
!  
interface FastEthernet0/21  
switchport access vlan 999  
switchport mode access  
shutdown  
!  
interface FastEthernet0/22  
no switchport  
no ip address  
channel-group 1 mode active  
duplex auto  
speed auto  
!  
interface FastEthernet0/23  
no switchport  
no ip address  
channel-group 1 mode active  
duplex auto  
speed auto  
!  
interface FastEthernet0/24  
no switchport  
no ip address  
channel-group 1 mode active  
duplex auto  
speed auto  
!  
interface GigabitEthernet0/1  
switchport trunk allowed vlan 10  
switchport trunk encapsulation dot1q  
switchport mode trunk  
!  
interface GigabitEthernet0/2  
switchport trunk allowed vlan 20  
switchport trunk encapsulation dot1q  
switchport mode trunk  
!
```

MLS3

```

interface Vlan1
no ip address
shutdown
!
interface Vlan10
mac-address 0007.ece5.7b01
ip address 192.168.10.2 255.255.255.224
ip helper-address 192.168.10.122
ip access-group GUEST_VLAN10_RESTRICTION in
standby 10 ip 192.168.10.1
standby 10 priority 110
standby 10 preempt
!
interface Vlan20
mac-address 0007.ece5.7b02
ip address 192.168.10.34 255.255.255.224
ip helper-address 192.168.10.122
standby 20 ip 192.168.10.33
standby 20 preempt
!
interface Vlan100
mac-address 0007.ece5.7b03
no ip address
ip helper-address 192.168.10.122
!
router ospf 1
log-adjacency-changes
passive-interface GigabitEthernet0/1
passive-interface GigabitEthernet0/2
network 1.1.1.1 0.0.0.0 area 0
network 192.168.10.0 0.0.0.31 area 0
network 192.168.10.32 0.0.0.31 area 0
network 192.168.10.112 0.0.0.3 area 0
network 192.168.10.128 0.0.0.3 area 0
network 192.168.10.132 0.0.0.3 area 0
network 192.168.10.140 0.0.0.3 area 0
network 192.168.10.124 0.0.0.3 area 0
!
ip classless
!
ip flow-export version 9
!
!

```

MLS3

```

ip access-list extended GUEST_VLAN10_RESTRICTION
permit udp any eq bootpc any eq bootps
permit tcp 192.168.10.0 0.0.0.31 host 192.168.10.154
deny ip 192.168.10.0 0.0.0.31 192.168.10.0 0.0.0.255
permit ip 192.168.10.0 0.0.0.31 any
ip access-list extended GUE
!
banner motd ^C
*****
WARNING: Authorized Access Only
This device is property of the School Network.
Unauthorized access is strictly prohibited.
*****
^C
!
!
!
!
!
line con 0
password 7 082048430017544541
logging synchronous
login local
!
line aux 0
!
line vty 0 4
login local
transport input ssh
line vty 5 15
login local
transport input ssh
!
!
!
!
end

```

MLS4

Current configuration : 5533 bytes

MLS4

```
!  
version 12.2(37)SE1  
no service timestamps log datetime msec  
no service timestamps debug datetime msec  
service password-encryption  
!  
hostname MLS4  
!  
!  
enable secret 5 $1$mERr$AFX/pZT1Lh7NP3Dp3P/qq/  
!  
!  
ip dhcp excluded-address 192.168.10.1 192.168.10.3  
ip dhcp excluded-address 192.168.10.33 192.168.10.35  
ip dhcp excluded-address 192.168.10.3  
ip dhcp excluded-address 192.168.10.35  
!  
ip dhcp pool VLAN10  
network 192.168.10.0 255.255.255.224  
default-router 192.168.10.1  
dns-server 8.8.8.8  
ip dhcp pool VLAN20  
network 192.168.10.32 255.255.255.224  
default-router 192.168.10.33  
dns-server 8.8.8.8  
!  
!  
ip routing  
!  
!  
!  
!  
username Admin secret 5 $1$mERr$AFX/pZT1Lh7NP3Dp3P/qq/  
!  
!  
!  
!  
!  
!  
!  
!  
!
```

MLS4

```
ip ssh version 2
ip ssh time-out 60
ip domain-name school.edu
!
!
spanning-tree mode pvst
!
!
!
!
!
!
!
interface Loopback2
ip address 2.2.2.2 255.255.255.255
!
interface Port-channel1
no switchport
ip address 192.168.10.114 255.255.255.252
!
interface Port-channel3
no switchport
ip address 192.168.10.121 255.255.255.252
!
interface FastEthernet0/1
no switchport
ip address 192.168.10.153 255.255.255.252
duplex auto
speed auto
!
interface FastEthernet0/2
no switchport
ip address 192.168.10.137 255.255.255.252
duplex auto
speed auto
!
interface FastEthernet0/3
switchport access vlan 999
switchport mode access
shutdown
!
interface FastEthernet0/4
switchport access vlan 999
switchport mode access
```

MLS4

```
shutdown
!  
interface FastEthernet0/5  
switchport access vlan 999  
switchport mode access  
shutdown  
!  
interface FastEthernet0/6  
switchport access vlan 999  
switchport mode access  
shutdown  
!  
interface FastEthernet0/7  
switchport access vlan 999  
switchport mode access  
shutdown  
!  
interface FastEthernet0/8  
switchport access vlan 999  
switchport mode access  
shutdown  
!  
interface FastEthernet0/9  
switchport access vlan 999  
switchport mode access  
shutdown  
!  
interface FastEthernet0/10  
switchport access vlan 999  
switchport mode access  
shutdown  
!  
interface FastEthernet0/11  
switchport access vlan 999  
switchport mode access  
shutdown  
!  
interface FastEthernet0/12  
switchport access vlan 999  
switchport mode access  
shutdown  
!  
interface FastEthernet0/13
```

MLS4

```
switchport access vlan 999
switchport mode access
shutdown
!
interface FastEthernet0/14
switchport access vlan 999
switchport mode access
shutdown
!
interface FastEthernet0/15
switchport access vlan 999
switchport mode access
shutdown
!
interface FastEthernet0/16
switchport access vlan 999
switchport mode access
shutdown
!
interface FastEthernet0/17
switchport access vlan 999
switchport mode access
shutdown
!
interface FastEthernet0/18
no switchport
ip address 192.168.10.133 255.255.255.252
duplex auto
speed auto
!
interface FastEthernet0/19
no switchport
no ip address
channel-group 3 mode active
duplex auto
speed auto
!
interface FastEthernet0/20
no switchport
no ip address
channel-group 3 mode active
duplex auto
speed auto
```

MLS4

```
!  
interface FastEthernet0/21  
no switchport  
no ip address  
channel-group 3 mode active  
duplex auto  
speed auto  
!  
interface FastEthernet0/22  
no switchport  
no ip address  
channel-group 1 mode active  
duplex auto  
speed auto  
!  
interface FastEthernet0/23  
no switchport  
no ip address  
channel-group 1 mode active  
duplex auto  
speed auto  
!  
interface FastEthernet0/24  
no switchport  
no ip address  
channel-group 1 mode active  
duplex auto  
speed auto  
!  
interface GigabitEthernet0/1  
switchport trunk allowed vlan 20  
switchport trunk encapsulation dot1q  
switchport mode trunk  
!  
interface GigabitEthernet0/2  
switchport trunk allowed vlan 10  
switchport trunk encapsulation dot1q  
switchport mode trunk  
!  
interface Vlan1  
no ip address  
shutdown  
!
```


MLS4

```
interface Vlan10
mac-address 0002.16ad.5d01
ip address 192.168.10.3 255.255.255.224
ip helper-address 192.168.10.122
ip access-group GUESS_VLAN10_RESTRICTION in
standby 10 ip 192.168.10.1
standby 10 preempt
!
interface Vlan20
mac-address 0002.16ad.5d02
ip address 192.168.10.35 255.255.255.224
ip helper-address 192.168.10.122
standby 20 ip 192.168.10.33
standby 20 priority 110
standby 20 preempt
!
interface Vlan100
mac-address 0002.16ad.5d03
no ip address
ip helper-address 192.168.10.122
!
router ospf 1
log-adjacency-changes
passive-interface GigabitEthernet0/1
passive-interface GigabitEthernet0/2
network 2.2.2.2 0.0.0.0 area 0
network 192.168.10.0 0.0.0.31 area 0
network 192.168.10.112 0.0.0.3 area 0
network 192.168.10.120 0.0.0.3 area 0
network 192.168.10.136 0.0.0.3 area 0
network 192.168.10.140 0.0.0.3 area 0
network 192.168.10.132 0.0.0.3 area 0
network 192.168.10.152 0.0.0.3 area 0
!
ip classless
!
ip flow-export version 9
!
!
ip access-list extended GUESS_VLAN10_RESTRICTION
permit udp any eq bootps any eq bootpc
permit tcp 192.168.10.0 0.0.0.31 host 192.168.10.154 eq ftp
permit tcp 192.168.10.0 0.0.0.31 host 192.168.10.154 eq 20
```

MLS4

```

deny ip 192.168.10.0 0.0.0.31 192.168.10.0 0.0.0.255
permit ip 192.168.10.0 0.0.0.31 any
ip access-list extended GUEST_VLAN10_RESTRICTION
permit tcp 192.168.10.0 0.0.0.31 host 192.168.10.154
!
banner motd ^C
*****
WARNING: Authorized Access Only
This device is property of the School Network.
Unauthorized access is strictly prohibited.
*****
^C
!
!
!
!
!
line con 0
password 7 082048430017544541
logging synchronous
login local
!
line aux 0
!
line vty 0 4
login local
transport input ssh
line vty 5 15
login local
transport input ssh
!
!
!
!
end

```

MLS5

```

Current configuration : 5533 bytes
!
version 12.2(37)SE1
no service timestamps log datetime msec
no service timestamps debug datetime msec
service password-encryption
!
hostname MLS5
!
enable secret 5 $1$mERr$AFX/pZT1Lh7NP3Dp3P/qq/
!
ip dhcp excluded-address 192.168.10.65 192.168.10.67
ip dhcp excluded-address 192.168.10.81 192.168.10.83
ip dhcp excluded-address 192.168.10.97 192.168.10.99
ip dhcp excluded-address 192.168.10.66
ip dhcp excluded-address 192.168.10.82
ip dhcp excluded-address 192.168.10.98
ip dhcp excluded-address 192.168.10.52
!
ip dhcp pool IT_DEP
network 192.168.10.64 255.255.255.240
default-router 192.168.10.65
dns-server 8.8.8.8
ip dhcp pool LAB
network 192.168.10.80 255.255.255.240
default-router 192.168.10.81
dns-server 8.8.8.8
ip dhcp pool HR
network 192.168.10.96 255.255.255.240
default-router 192.168.10.97
dns-server 8.8.8.8
!
!
ip routing
!
username Admin secret 5 $1$mERr$AFX/pZT1Lh7NP3Dp3P/qq/
!
ip ssh version 2
ip ssh time-out 60
ip domain-name school.edu
!
!
spanning-tree mode pvst

```

MLS5

```
!  
interface Loopback3  
ip address 3.3.3.3 255.255.255.255  
!  
interface Port-channel1  
switchport access vlan 999  
switchport mode access  
shutdown  
!  
interface Port-channel2  
no switchport  
ip address 192.168.10.117 255.255.255.252  
!  
interface Port-channel3  
no switchport  
ip address 192.168.10.122 255.255.255.252  
!  
interface FastEthernet0/1  
no switchport  
ip address 192.168.10.169 255.255.255.252  
duplex auto  
speed auto  
!  
interface FastEthernet0/2  
switchport access vlan 999  
switchport mode access  
shutdown  
!  
interface FastEthernet0/3  
no switchport  
ip address 192.168.10.145 255.255.255.252  
duplex auto  
speed auto  
!  
interface FastEthernet0/4  
no switchport  
ip address 192.168.10.149 255.255.255.252  
duplex auto  
speed auto  
!  
interface FastEthernet0/5  
switchport access vlan 999  
switchport mode access
```

MLS5

```
shutdown
!  
interface FastEthernet0/6  
switchport access vlan 999  
switchport mode access  
shutdown  
!  
interface FastEthernet0/7  
switchport access vlan 999  
switchport mode access  
shutdown  
!  
interface FastEthernet0/8  
switchport access vlan 999  
switchport mode access  
shutdown  
!  
interface FastEthernet0/9  
switchport access vlan 999  
switchport mode access  
shutdown  
!  
interface FastEthernet0/10  
no switchport  
no ip address  
duplex auto  
speed auto  
!  
interface FastEthernet0/11  
switchport access vlan 999  
switchport mode access  
shutdown  
!  
interface FastEthernet0/12  
switchport access vlan 999  
switchport mode access  
shutdown  
!  
interface FastEthernet0/13  
switchport access vlan 999  
switchport mode access  
shutdown  
!
```

MLS5

```
interface FastEthernet0/14
switchport access vlan 999
switchport mode access
shutdown
!
interface FastEthernet0/15
switchport access vlan 999
switchport mode access
shutdown
!
interface FastEthernet0/16
switchport access vlan 999
switchport mode access
shutdown
!
interface FastEthernet0/17
no switchport
ip address 192.168.10.126 255.255.255.252
duplex auto
speed auto
!
interface FastEthernet0/18
switchport access vlan 999
switchport mode access
shutdown
!
interface FastEthernet0/19
no switchport
no ip address
channel-group 3 mode active
duplex auto
speed auto
!
interface FastEthernet0/20
no switchport
no ip address
channel-group 3 mode active
duplex auto
speed auto
!
interface FastEthernet0/21
no switchport
no ip address
```

MLS5

```
channel-group 3 mode active
duplex auto
speed auto
!
interface FastEthernet0/22
switchport trunk encapsulation dot1q
channel-group 2 mode active
!
interface FastEthernet0/23
switchport trunk encapsulation dot1q
channel-group 2 mode active
!
interface FastEthernet0/24
switchport trunk encapsulation dot1q
channel-group 2 mode active
!
interface GigabitEthernet0/1
switchport trunk allowed vlan 30,40
switchport trunk encapsulation dot1q
switchport mode trunk
!
interface GigabitEthernet0/2
switchport trunk allowed vlan 50,100
switchport trunk encapsulation dot1q
switchport mode trunk
!
interface Vlan1
no ip address
shutdown
!
interface Vlan30
mac-address 0060.2f3d.c401
ip address 192.168.10.66 255.255.255.240
ip helper-address 192.168.10.126
standby 30 ip 192.168.10.65
standby 30 priority 110
standby 30 preempt
!
interface Vlan40
mac-address 0060.2f3d.c402
ip address 192.168.10.82 255.255.255.240
ip helper-address 192.168.10.126
standby 40 ip 192.168.10.81
```

MLS5

```

standby 40 priority 110
standby 40 preempt
!
interface Vlan50
mac-address 0060.2f3d.c403
ip address 192.168.10.98 255.255.255.240
ip helper-address 192.168.10.126
standby 50 ip 192.168.10.97
standby 50 preempt
!
interface Vlan100
mac-address 0060.2f3d.c404
ip address 192.168.10.52 255.255.255.240
!
router ospf 1
log-adjacency-changes
passive-interface GigabitEthernet0/1
passive-interface GigabitEthernet0/2
network 3.3.3.3 0.0.0.0 area 0
network 192.168.10.48 0.0.0.15 area 0
network 192.168.10.64 0.0.0.15 area 0
network 192.168.10.80 0.0.0.15 area 0
network 192.168.10.96 0.0.0.15 area 0
network 192.168.10.116 0.0.0.3 area 0
network 192.168.10.144 0.0.0.3 area 0
network 192.168.10.148 0.0.0.3 area 0
network 192.168.10.124 0.0.0.3 area 0
network 192.168.10.120 0.0.0.3 area 0
!
ip classless
!
ip flow-export version 9
!
!
!
banner motd ^C
*****
WARNING: Authorized Access Only
This device is property of the School Network.
Unauthorized access is strictly prohibited.
*****
^C
!
```


MLS5

```

line con 0
password 7 082048430017544541
logging synchronous
login local
!
line aux 0
!
line vty 0 4
login local
transport input ssh
line vty 5 15
login local
transport input ssh
!
end

```

MLS6

```

Current configuration : 5533 bytes
!
version 12.2(37)SE1
no service timestamps log datetime msec
no service timestamps debug datetime msec
service password-encryption
!
hostname MLS6
!
enable secret 5 $1$mERr$AFX/pZT1Lh7NP3Dp3P/qq/
!
ip dhcp excluded-address 192.168.10.65 192.168.10.67
ip dhcp excluded-address 192.168.10.81 192.168.10.83
ip dhcp excluded-address 192.168.10.97 192.168.10.99
ip dhcp excluded-address 192.168.10.67
ip dhcp excluded-address 192.168.10.83
ip dhcp excluded-address 192.168.10.99
ip dhcp excluded-address 192.168.10.53
!
ip dhcp pool IT_DEP
network 192.168.10.64 255.255.255.240
default-router 192.168.10.65
dns-server 8.8.8.8

```

MLS6

```
ip dhcp pool LAB
network 192.168.10.80 255.255.255.240
default-router 192.168.10.81
dns-server 8.8.8.8
ip dhcp pool HR
network 192.168.10.96 255.255.255.240
default-router 192.168.10.97
dns-server 8.8.8.8
!
!
ip routing
!
username Admin secret 5 $1$mERr$AFX/pZT1Lh7NP3Dp3P/qq/
!
ip ssh version 2
ip ssh time-out 60
ip domain-name school.edu
!
!
spanning-tree mode pvst
!
interface Loopback4
ip address 4.4.4.4 255.255.255.255
!
interface Port-channel2
no switchport
ip address 192.168.10.118 255.255.255.252
!
interface FastEthernet0/1
switchport access vlan 999
switchport mode access
shutdown
!
interface FastEthernet0/2
switchport access vlan 999
switchport mode access
shutdown
!
interface FastEthernet0/3
no switchport
ip address 192.168.10.157 255.255.255.252
duplex auto
speed auto
```

MLS6

```
!  
interface FastEthernet0/4  
no switchport  
ip address 192.168.10.153 255.255.255.252  
duplex auto  
speed auto  
!  
interface FastEthernet0/5  
switchport access vlan 999  
switchport mode access  
shutdown  
!  
interface FastEthernet0/6  
switchport access vlan 999  
switchport mode access  
shutdown  
!  
interface FastEthernet0/7  
switchport access vlan 999  
switchport mode access  
shutdown  
!  
interface FastEthernet0/8  
switchport access vlan 999  
switchport mode access  
shutdown  
!  
interface FastEthernet0/9  
switchport access vlan 999  
switchport mode access  
shutdown  
!  
interface FastEthernet0/10  
switchport access vlan 999  
switchport mode access  
shutdown  
!  
interface FastEthernet0/11  
switchport access vlan 999  
switchport mode access  
shutdown  
!  
interface FastEthernet0/12
```

MLS6

```
switchport access vlan 999
switchport mode access
shutdown
!
interface FastEthernet0/13
switchport access vlan 999
switchport mode access
shutdown
!
interface FastEthernet0/14
switchport access vlan 999
switchport mode access
shutdown
!
interface FastEthernet0/15
switchport access vlan 999
switchport mode access
shutdown
!
interface FastEthernet0/16
switchport access vlan 999
switchport mode access
shutdown
!
interface FastEthernet0/17
switchport access vlan 999
switchport mode access
shutdown
!
interface FastEthernet0/18
no switchport
ip address 192.168.10.134 255.255.255.252
duplex auto
speed auto
!
interface FastEthernet0/19
switchport access vlan 999
switchport mode access
shutdown
!
interface FastEthernet0/20
no switchport
ip address 192.168.10.142 255.255.255.252
```

MLS6

```
duplex auto
speed auto
!
interface FastEthernet0/21
switchport access vlan 999
switchport mode access
shutdown
!
interface FastEthernet0/22
switchport trunk encapsulation dot1q
channel-group 2 mode active
!
interface FastEthernet0/23
switchport trunk encapsulation dot1q
channel-group 2 mode active
!
interface FastEthernet0/24
switchport trunk encapsulation dot1q
channel-group 2 mode active
!
interface GigabitEthernet0/1
switchport trunk allowed vlan 50,100
switchport trunk encapsulation dot1q
switchport mode trunk
!
interface GigabitEthernet0/2
switchport trunk allowed vlan 30,40
switchport trunk encapsulation dot1q
switchport mode trunk
!
interface Vlan1
no ip address
shutdown
!
interface Vlan30
mac-address 0002.4a92.e901
ip address 192.168.10.67 255.255.255.240
ip helper-address 192.168.10.126
standby 30 ip 192.168.10.65
standby 30 priority 90
standby 30 preempt
!
interface Vlan40
```

MLS6

```

mac-address 0002.4a92.e902
ip address 192.168.10.83 255.255.255.240
ip helper-address 192.168.10.126
standby 40 ip 192.168.10.81
standby 40 priority 90
standby 40 preempt
!
interface Vlan50
mac-address 0002.4a92.e903
ip address 192.168.10.99 255.255.255.240
ip helper-address 192.168.10.126
standby 50 ip 192.168.10.97
standby 50 priority 110
standby 50 preempt
!
interface Vlan100
mac-address 0002.4a92.e904
ip address 192.168.10.53 255.255.255.240
!
router ospf 1
log-adjacency-changes
passive-interface GigabitEthernet0/1
passive-interface GigabitEthernet0/2
network 4.4.4.4 0.0.0.0 area 0
network 192.168.10.48 0.0.0.15 area 0
network 192.168.10.64 0.0.0.15 area 0
network 192.168.10.80 0.0.0.15 area 0
network 192.168.10.96 0.0.0.15 area 0
network 192.168.10.116 0.0.0.3 area 0
network 192.168.10.152 0.0.0.3 area 0
network 192.168.10.156 0.0.0.3 area 0
network 192.168.10.140 0.0.0.3 area 0
network 192.168.10.132 0.0.0.3 area 0
!
ip classless
!
ip flow-export version 9
!
banner motd ^C
*****

WARNING: Authorized Access Only
This device is property of the School Network.
Unauthorized access is strictly prohibited.

```

MLS6

^C

!

line con 0

password 7 082048430017544541

logging synchronous

login local

!

line aux 0

!

line vty 0 4

login local

transport input ssh

line vty 5 15

login local

transport input ssh

!

end